



Test Plan:

The Planets Solar System (Application)

Aim of Test Plan File: This is to verify that the console application accurately stores, displays, and queries planetary data while maintaining a stable user interface

1. Unit Testing

Data integrity

- **Test Case 1a: Data Loading**
 - **Action:** Launch the program.
 - **Expected Result:** The **PlanetData** class should successfully load all 10 objects which are from Mercury through Pluto into the list.
- **Test Case 1b: Moon Counts**
 - **Action:** Check the moon list for Mars
 - **Expected Result:** The system should return exactly 2 moons which are Phobos, Deimos

2. Functional Testing

User Features

- **Test Case 2a: Search Functionality**
 - **Action:** Select option (2) and type Jupiter
 - **Expected Result:** The console displays Jupiter's mass, distance, and moon count in its **DarkYellow** color
- **Test Case 2b: Existence Check**
 - **Action:** Select option (4) and type Pluto
 - **Expected Result:** The system confirms "Yes, it is in our system" even though Pluto's planetary status is often debated, I personally believe that it is a planet 😊

3. UI

Boundary Testing

- **Test Case 3a: Side by Side Display**
 - **Action:** Select option (1)
 - **Expected Result:** Planets are displayed in two neat columns using string padding to prevent any line break errors or crashes from occurring
- **Test Case 3b: Case Sensitivity**

- **Action:** Search for earth, (make sure in lowercase !)
- **Expected Result:** The `OrdinalIgnoreCase` logic should successfully find the planet regardless of whether or not there are capital letters present

4. Error Handling & Validation

Queries

- **Test Case 4a: Invalid Menu Input**
 - **Action:** Enter "9" or "Hello" in the main menu
 - **Expected Result:** The `ShowError` method triggers and will display a **red** error message without crashing the program
- **Test Case 4b: Null or Empty Search**
 - **Action:** Press **ENTER** without typing anything in the search menu
 - **Expected Result:** The program should handle the empty string efficiently and return to the menu

AI Transparency: Ai was not used for any part of this assignment , everything was done with my own knowledge/Understanding and learning materials/extra learning materials